Hai Yu

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Education

Nankai University Tianjin, China
Ph.D. in Control Science and Engineering; ***GPA: 88.06/100*** *Sep 2020 – Jun 2026 (expected)*
Supervisor: Assoc. Prof. ***Xiao Liang****, Prof.* ***Yongchun Fang****, and Prof.* ***Jianda Han***

Jilin University Changchun, China
B.Eng. in Automation; ***GPA: 90.91/100*** *Sep 2016 – Jun 2020*

Research Interests

Motion control and planning for autonomous systems, particularly aerial transportation and manipulation platforms.

Grants & Fellowships

NSFC Youth Student Basic Research Project (for Ph.D. students)
Principal Investigator *May 2024 – Dec 2025*

- Project Title: Research on Control Methods for Precise Release of Cargo with a Dual Underactuated Aerial Transportation System onto Mobile Vessels.
- Amount: **CNY 300,000**.
- Only about 500 positions nationwide each year in China.

CAST Youth Talent Support Project Doctoral Special Program
Principal Investigator *Jan 2025 – Jun 2026*

- Mentored and supported under the guidance of the Chinese Society of Aeronautics and Astronautics
- Amount: **CNY 40,000**.
- 3,226 scholars selected in the first nationwide cohort in China.

Awards & Honors

National Scholarship of China: Top national scholarship awarded by the Ministry of Education of China to the top 2% of students in the university. (Nov 2018, Nov 2024, Nov 2025)

Tianjin Outstanding Student: Municipal-level honor for top students in Tianjin. (Nov 2024)

Zhou Enlai Scholarship: Nankai university's highest student honor, awarded to *only 10 students per year*. (Jun 2025)

Nankai "Student of the Year" - Nomination: Nominee for Nankai University's top annual student honor. (Nov 2024)

Nankai University First Prize Graduate Scholarship: First-class graduate scholarship for outstanding research performance. (Nov 2023, Nov 2024, Nov 2025)

Nankai University May Fourth Youth Medal - "Youth Leader": University-level award for student leaders with major contributions to campus life. (May 2025)

First Prize, National Postdoctoral Academic Forum on "Xinchuang" Development: First prize for a research presentation at a national academic forum on innovation in information technology applications. (Dec 2022)

First Prize, North China Five-Province Collegiate Robotics Competition: First place in a regional collegiate robotics contest among universities from five northern provinces in China. (Nov 2024)

First Prize, Tianjin Intellectual Property Innovation & Entrepreneurship - Invention & Design Competition: First prize in a municipal-level competition on invention, design, and innovation-driven entrepreneurship. (Dec 2024)

Second Prize, International Underwater Robot Competition: Second prize in an international competition on underwater robotics. (Jul 2018)

Publications

Co-Authors marked with # contributed equally.

Aerial Transportation Systems:

** Position tracking, payload landing, visual servoing, and trajectory-refinement control for aerial transportation systems with variable-length cable:*

1. [**TII'2024**] **Hai Yu**, Xiao Liang, Jianda Han, Yongchun Fang, Adaptive Trajectory Tracking Control for the Quadrotor Aerial Transportation System Landing a Payload Onto the Mobile Platform, *IEEE Transactions on Industrial Informatics*, 2024, 20(1): 23-37. [PDF]

2. [**T-ASE'2025**] **Hai Yu**, Zhaopeng Zhang, Tengfei Pei, Jianda Han, Yongchun Fang, Xiao Liang, Visual Servoing-Based Anti-Swing Control of Cable-Suspended Aerial Transportation Systems With Variable-Length Cable, *IEEE Transactions on Automation Science and Engineering*, 2025, 22: 5955-5965. [PDF]

3. [**IROS'2025**] **Hai Yu**, Zhichao Yang, Wei He, Jianda Han, Yongchun Fang, Xiao Liang, Online Anti-Swing Trajectory Refinement for Variable-Length Cable-Suspended Aerial Transportation Robot, *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems.* [to appear]

4. [**TIE'2023**] Xiao Liang, **Hai Yu**, Zhuang Zhang, Huiying Ye, Yongchun Fang, Jianda Han, Unmanned Aerial Transportation System With Flexible Connection Between the Quadrotor and the Payload: Modeling, Controller Design, and Experimental Validation, *IEEE Transactions on Industrial Electronics*, 2023, 70(2): 1870-1882. [PDF]

** Modeling, control, and fault-tolerant scheme for aerial transportation systems with rotor blade damage:*

5. [**TIE'2024**] **Hai Yu**, Shizhen Wu, Wei He, Xiao Liang, Jianda Han, Yongchun Fang, Fault-Tolerant Control for Multirotor Aerial Transportation Systems With Blade Damage, *IEEE Transactions on Industrial Electronics*, 2024, 71(10): 12718-12731. [PDF]

** Anti-swing control frameworks considering hook dynamics, fully actuated UAVs, and cooperative multi-UAV transportation:*

6. [**TFS'2025**] **Hai Yu**, Yi Chai, Zhichao Yang, Jianda Han, Yongchun Fang, Xiao Liang, Fuzzy-Based Antiswing Control for Variable-Length Cable-Suspended Aerial Transportation Systems Considering the Hook Effect, *IEEE Transactions on Fuzzy Systems*, 2025, 33(2): 621-630. [PDF]

7. [**T-Mech'2024**] Kexin Cai#, **Hai Yu#**, Wei He, Xiao Liang, Jianda Han, Yongchun Fang, An Enhanced-Coupling Control Method for Aerial Transportation Systems With Double-Pendulum Swing Effects, *IEEE/ASME Transactions on Mechatronics*, 2024, 29(3):2302-2315. [PDF]

8. [**TII'2025**] Zhuang Zhang#, **Hai Yu#**, Huiying Ye, Jianda Han, Yongchun Fang, Xiao Liang, Collaborative Control for Aerial Transportation of Cargo With Dual Quadrotors, *IEEE Transactions on Industrial Informatics*, 2025, 21(1): 752-761. [PDF]

9. [**TIE'2023**] Zhichao Yang, **Hai Yu**, Yi Chai, Wei He, Xiao Liang, Jianda Han, Omni-Directional Aerial Transportation System: Modeling, Control, and Experimental Validation, *IEEE Transactions on Industrial Electronics*, 2025, 72(8): 8261-8269. [PDF]

Aerial Manipulation Systems:

** Motion planning and robust nonlinear control methods for single- and dual-arm aerial manipulators operating in constrained environments and under strong coupling:*

10. [**T-Mech'2025**] Zhaopeng Zhang, **Hai Yu**, Yi Chai, Zhichao Yang, Xiao Liang, Yongchun Fang, Jianda Han, An End-Effector-Oriented Coupled Motion Planning Method for Aerial Manipulators in Constrained Environments, *IEEE/ASME Transactions on Mechatronics*, DOI: 10.1109/TMECH.2025.3550562. [PDF]

11. [**CEP'2025**] Kexin Cai, **Hai Yu**, Zhaopeng Zhang, Xiao Liang, Yongchun Fang, Jianda Han, An Experiment Study for Unmanned Aerial Manipulator Systems With L1 Adaptive Augmentation of Geometric Control, *Control Engineering Practice*, 2025, 164: 106418. [PDF]

12. [**GNC'2023**] Bingbing Liu, **Hai Yu**, Shizhen Wu, Xiao Liang, Yongchun Fang, Adaptive Sliding-Mode Disturbance Observer-Based Nonlinear Control for Unmanned Dual-Arm Aerial Manipulator Subject to State Constraints, *Guidance, Navigation and Control*, 2023, 3(3): 2350021. [PDF]

13. [**T-ASE'2025**] Xiao Liang, Yang Wang, **Hai Yu**, Zhaopeng Zhang, Jianda Han, Yongchun Fang, Observer-Based Nonlinear Control for Dual-Arm Aerial Manipulator Systems Suffering From Uncertain Center of Mass, *Guidance, Navigation and Control*, 2024, 22: 1984-1995. [PDF]

Blimps:

** Trajectory tracking control frameworks for indoor unmanned blimps under disturbances and measurement noise, from controller design to hardware validation:*

14. [**TIE'2025**] Jinyang Dong, **Hai Yu**, Biao Lu, Huawang Liu, Yongchun Fang, Adaptive Output Feedback Trajectory Tracking Control of an Indoor Blimp: Controller Design and Experiment Validation, *IEEE Transactions on Industrial Electronics*, 2025, 72(4): 3960-3971. [PDF]

15. [**CEP'2025**] Jinyang Dong, Honglin Han, **Hai Yu**, Biao Lu, Xiao Liang, Yongchun Fang, Finite Time Tracking Control of a Miniature Unmanned Blimp with Disturbances and Measurement Noises: A Neural Network-Based Approach, *Control Engineering Practice*, 2025, 165: 106539. [PDF]

Skills

Languages: Chinese (native), English (Advanced)

Coding: Python, Matlab and LaTeX

Control: Nonlinear and safety-critical control of underactuated aerial transportation and manipulation systems

Software: Frequent use of ROS1, Simulink, Adobe Illustrator and Office

Hardware: Proficient in NVIDIA Jetson, Raspberry Pi, Qualisys motion capture system and Intel RealSense cameras

Academic Services

IEEE Student Member

IEEE Robotics and Automation Society Student Member

IEEE Industrial Electronics Society Student Member

IEEE Control Systems Society Student Member

Chinese Society of Aeronautics and Astronautics Student Member

Chinese Association of Automation Student Member

Journal/Conference Reviewers:

- IEEE T-RO, IEEE/ASME T-Mech, IEEE TII, IEEE RA-L, IEEE CS-L, AST, IEEE CDC, IEEE/RSJ IROS